Journal of Computational Physics 295 (2015) 65–86

Contents lists available at ScienceDirect

Journal of Computational Physics

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DG-FTLE: Lagrangian coherent structures with high-order discontinuous-Galerkin methods

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ARTICLE INFO

Article history:
Received 7 May 2014
Received in revised form 13 January 2015
Accepted 17 March 2015
Available online 9 April 2015

Keywords:
Lagrangian coherent structures
Finite-time Lyapunov exponent
Discontinuous-Galerkin
Spectral element method
Higher-order

ABSTRACT

We present an algorithm for the computation of finite-time Lyapunov exponent (FTLE) fields using discontinuous-Galerkin (dG) methods in two dimensions. The algorithm is designed to compute FTLE fields simultaneously with the time integration of dG-based flow solvers of conservation laws. Fluid tracers are initialized at Gauss–Lobatto quadrature nodes within an element. The deformation gradient tensor, defined by the deformation of the Lagrangian flow map in finite time, is determined per element with high-order dG operators. Multiple flow maps are constructed from a particle trace that is released at a single initial time by mapping and interpolating the flow map formed by the locations of the fluid tracers after finite time integration to a unit square master element and to the quadrature nodes within the element, respectively. The interpolated flow maps are used to compute forward-time and backward-time FTLE fields at several times using dG operators. For a large finite integration time, the interpolation is increasingly poorly conditioned because of the excessive subdomain deformation. The conditioning can be used in addition to the FTLE to quantify the deformation of the flow field and identify subdomains with material lines that define Lagrangian coherent structures. The algorithm is tested on three benchmarks: an analytical spatially periodic gyre flow, a vortex advected by a uniform inviscid flow, and the viscous flow around a square cylinder. In these cases, the algorithm is shown to have spectral convergence.

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1. Introduction

The Lagrangian coherent structure (LCS) theory proposed by Haller and Yuan [1] and Haller [2] has evolved as a valuable compliment to Eulerian visualization methods, such as the Q, λ_2 and Δ criteria [3,4], in the elucidation of flow topology. Haller [5] defines LCSs as hyperbolic material lines or surfaces which exhibit locally the strongest attraction or repulsion in a flow. Using a variational approach, criteria can be constructed to ensure hyperbolicity in material lines from the eigenvector fields of the Cauchy–Green strain tensor. Although very precise, Farazmand and Haller [6] show that the numerical determination of LCSs in this approach is computationally intensive. The variational definition of LCSs is generalized in the work of Haller and Beron-Vera [7] to include parabolic and elliptic LCS defined by minimal geodesics of the Riemannian metric induced by the Cauchy–Green strain tensor. Shadden et al. [8] propose an alternative LCS definition in two dimensions in which repelling fluid barriers are identified by ridges in the finite-time Lyapunov exponent (FTLE) field computed

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http://dx.doi.org/10.1016/j.jcp.2015.03.040
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forward in time, and attracting barriers are identified by ridges in the FTLE field computed backward in time. Lekien et al. [9] extended LCS definition to higher dimensions.

LCS methods have proven to be useful in the analysis of a range of applications involving transport phenomena (*e.g.* [10–14]). A drawback of LCS visualization is the high computational cost that is associated with the determination of the FTLE fields which requires particle tracing, and the determination of Jacobians in the particle deformation field and its associated eigenvalues. Moreover, if the LCS is post-processed from an unsteady flow computation, the time dependent velocity data in the entire domain must typically be stored, which is demanding on computer storage hardware. As a result, low-cost and low accuracy (typically finite difference) methods applied on low resolution datasets have been predominant for FTLE determination in large scale computational cases of unsteady flow in complex geometry.

In the past years, improving accuracy and efficiency of numerical LCS identification has received some attention. Garth et al. [15] use an adaptive particle sampling method to achieve higher resolution near FTLE ridges, thus reducing the overall number of required fluid tracers. Lekien and Ross [16] extend LCS theory to general Riemannian manifolds and introduce an efficient and accurate adaptive mesh algorithm for computing the FTLE. Brunton and Rowley [17] present an algorithm for approximating the flow map using low-order interpolation, eliminating redundant particle integrations in time-varying flows and improving efficiency in FTLE determination. Salman et al. [18] compute low-order accurate FTLE fields with finite-difference methods from velocity fields determined by high-order discontinuous-Galerkin (dG) methods. The requirement for high-fidelity resolution of ridges in the FTLE field is important, particularly in the area of flow control, as pointed out by Salman et al. [18]. In [19], we compute the deformation gradient tensor using dG operators, improving the accuracy of FTLE determination.

In this paper, we present a dG-based algorithm that determines flow maps and corresponding FTLE fields at multiple times from a particle trace at a single initial time, both backward in time and forward in time in two dimensions. Since the algorithm requires only a single particle trace at one initial time to determine the FTLE fields at several (forward and backward) times, it is low-storage and computationally efficient. By initiating particles at quadrature points, flow map approximations are generated that are consistent with the piecewise dG approximation. Hence, dG operators can be directly used to determine the deformation gradients required for the FTLE. Because of its seamless fit with dG Navier–Stokes solvers, the algorithm is not only consistent, but is also suited for the simultaneous computation with large-scale, parallel direct numerical simulations and large eddy simulations based on dG [20–23], at low computational overhead, while eliminating the need for storage of large data sets.

Although effort has been made to decrease storage requirements and improve efficiency of the presented algorithm as compared to currently available methods, the focus of this paper is on the high-order accuracy of the algorithm. Computational costs will be examined in detail elsewhere.

In the next section, we summarize a general dG flow solver, with a focus on the differential operators and mappings used in the FTLE algorithm. Then the LCS theory and the FTLE is discussed, followed by a presentation of the particle tracer, the flow map approximation and FTLE determination with dG. Next, the interpolation of flow maps at multiple times from a single initial time particle trace is discussed, including its conditioning and the relation of conditioning to the deformation field. The algorithm is tested on three benchmark cases, including a spatially periodic gyre flow, an analytical vortex advected by a uniform flow, and the viscous flow around a square cylinder. The final section is reserved for conclusions and future work.

2. Discontinuous-Galerkin Navier–Stokes solver

2.1. Governing equations

We consider two-dimensional flows that are governed by the following type of conservation laws:

$$\mathbf{Q}_t + \nabla \cdot \mathbf{F} = 0, \tag{1}$$

where the flux tensor, $\mathbf{F}$, can be separated into advective (a) and viscous (v) fluxes, such that

$$\nabla \cdot \mathbf{F} = \mathbf{F}^a_x + \mathbf{F}^a_y - \frac{1}{\mathrm{Re}_f}(\mathbf{F}^v_x + \mathbf{F}^v_y). \tag{2}$$

The subscripts x and y denote the spatial derivatives of the fluxes and Re_f is the Reynolds number.

2.2. Discontinuous-Galerkin method

The Navier–Stokes equations are discretized with a dG spectral element method. Here, we briefly summarize the numerical methodology with an emphasis on the topics that are pertinent to the determination of the FTLE. For detailed descriptions of dG methods, we refer to [24–26]. We begin by dividing the computational domain, Ω, into a set of non-overlapping quadrilateral subdomains, D_k, such that

$$\Omega = \bigcup_{k=1}^{K} D_k. \tag{3}$$

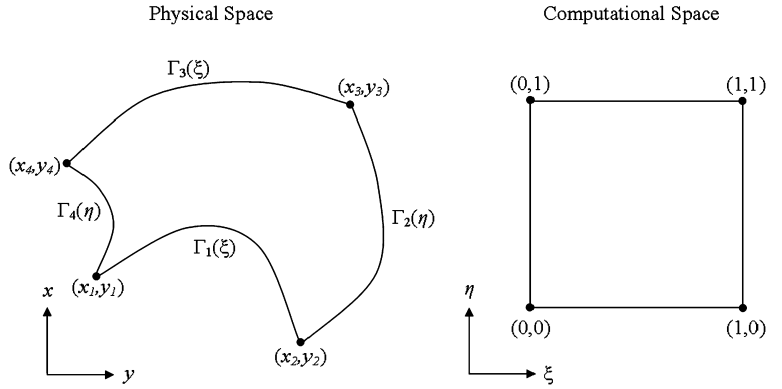


Fig. 1. Mapping from physical space (x, y) to the unit square in computational coordinates (ξ, η). The mapping is uniquely defined by the boundary polynomial representation Γ_i and the corner coordinates $\mathbf{x}_i$.

The subdomains, D_k, are mapped to the unit square as in Fig. 1 using a transfinite map given by the linear blending formula

$$\begin{aligned}\mathbf{x}(\xi,\eta) &= (1-\eta)\Gamma_1(\xi) + \eta\Gamma_3(\xi) + (1-\xi)\Gamma_4(\eta) + \xi\Gamma_2(\eta) \\ &\quad - \mathbf{x}_1(1-\xi)(1-\eta) - \mathbf{x}_2\xi(1-\eta) - \mathbf{x}_3\xi\eta - \mathbf{x}_4(1-\xi)\eta. \end{aligned} \tag{4}$$

The functions Γ_i are parametric curves that bound the subdomains. Under this mapping, the governing equation (1) becomes

$$\tilde{\mathbf{Q}}_t + \nabla\cdot\tilde{\mathbf{F}} = 0, \tag{5}$$

where,

$$\begin{aligned}\tilde{\mathbf{Q}} &= J\mathbf{Q}, \\ \tilde{\mathbf{F}}_i &= (\mathbf{a}_j \times \mathbf{a}_k)_i \cdot \mathbf{F}. \end{aligned} \tag{6}$$

In two dimensions, the vectors $\mathbf{a}_i$ represent the metric terms and are given by $\mathbf{a}_1 = \partial\mathbf{x}/\partial\xi$ and $\mathbf{a}_2 = \partial\mathbf{x}/\partial\eta$, where we have

$$\frac{\partial\mathbf{x}}{\partial\xi} = \mathbf{x}_1(1-\eta) - \mathbf{x}_2(1-\eta) - \mathbf{x}_3\eta + \mathbf{x}_4\eta + \Gamma_1'(1-\eta) + \Gamma_3'\eta + \Gamma_2 - \Gamma_4, \tag{7}$$

$$\frac{\partial\mathbf{x}}{\partial\eta} = \mathbf{x}_1(1-\xi) + \mathbf{x}_2\xi - \mathbf{x}_3\xi - \mathbf{x}_4(1-\xi) + \Gamma_2'\xi + \Gamma_4'(1-\xi) - \Gamma_1 + \Gamma_3. \tag{8}$$

Eqs. (7) and (8) form a set of covariant basis vectors as depicted in Fig. 2. The J in (6) is the Jacobian of the mapping,

$$J = \frac{\partial x}{\partial\xi}\frac{\partial y}{\partial\eta} - \frac{\partial y}{\partial\xi}\frac{\partial x}{\partial\eta}. \tag{9}$$

Gradients under the mapped subdomains are computed using the chain rule such that $\nabla_\xi F(\mathbf{x})$ is given by

$$\begin{pmatrix}\frac{\partial F}{\partial x}\\ \frac{\partial F}{\partial y}\end{pmatrix} = \begin{pmatrix}\frac{\partial\xi}{\partial x} & \frac{\partial\eta}{\partial x}\\ \frac{\partial\xi}{\partial y} & \frac{\partial\eta}{\partial y}\end{pmatrix}\begin{pmatrix}\frac{\partial F}{\partial\xi}\\ \frac{\partial F}{\partial\eta}\end{pmatrix} = (\mathbf{J}^T)^{-1}\begin{pmatrix}\frac{\partial F}{\partial\xi}\\ \frac{\partial F}{\partial\eta}\end{pmatrix}, \tag{10}$$

where $(\mathbf{J}^T)^{-1}$ is the inverse of the Jacobian transpose matrix $\mathbf{J}^T$ such that

$$(\mathbf{J}^T)^{-1} = \begin{pmatrix}\frac{\partial x}{\partial\xi} & \frac{\partial y}{\partial\xi}\\ \frac{\partial x}{\partial\eta} & \frac{\partial y}{\partial\eta}\end{pmatrix}^{-1} = \frac{1}{|\mathbf{J}^T|}\begin{pmatrix}\frac{\partial y}{\partial\eta} & -\frac{\partial y}{\partial\xi}\\ -\frac{\partial x}{\partial\eta} & \frac{\partial x}{\partial\xi}\end{pmatrix}, \tag{11}$$

where $|\mathbf{J}^T| = J$ as in (9). Combining (10) and (11) we have

$$\begin{aligned}\frac{\partial F}{\partial x} &= \frac{1}{J}\left[\frac{\partial F}{\partial\xi}\frac{\partial y}{\partial\eta} - \frac{\partial F}{\partial\eta}\frac{\partial y}{\partial\xi}\right], \\ \frac{\partial F}{\partial y} &= \frac{1}{J}\left[\frac{\partial F}{\partial\eta}\frac{\partial x}{\partial\xi} - \frac{\partial F}{\partial\xi}\frac{\partial x}{\partial\eta}\right]. \end{aligned} \tag{12}$$

$\tilde{\mathbf{Q}}$ is approximated with Nth-order Legendre polynomials in the computational coordinates, (ξ, η),

$$\tilde{\mathbf{Q}}_N(\xi,\eta,t) = \sum_{i=0}^{N}\sum_{j=0}^{N}\tilde{\mathbf{Q}}_{i,j}(t)\ell_i(\xi)\ell_j(\eta), \tag{13}$$

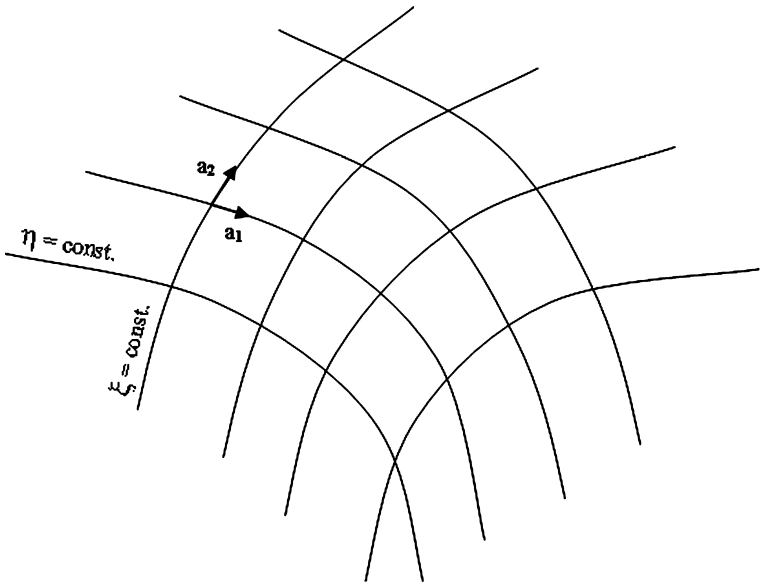


Fig. 2. Covariant bases definition.

where $\ell_i(\xi)$ and $\ell_j(\eta)$ are the Lagrange polynomials are given by

$$\ell_i(\xi) = \prod_{\substack{k=0 \\ k\neq i}}^{N} \frac{\xi - \xi_k}{\xi_i - \xi_k}. \tag{14}$$

The spatial derivative is determined as follows,

$$f'(\xi_i) \approx \sum_{j=0}^{N} f_j \ell'_j(\xi_i) = \sum_{j=0}^{N} D_{ij} f_j, \tag{15}$$

with D_{ij} given by

$$D_{ij} = \begin{cases} \dfrac{1}{\xi_j - \xi_i} \displaystyle\prod_{k=0}^{N} \frac{\xi_i - \xi_k}{\xi_j - \xi_k}, & i \neq j \\ \displaystyle\sum_{l=0}^{N} \frac{1}{\xi_j - \xi_l}, & i = j. \end{cases} \tag{16}$$

In the Galerkin approach, the residual is minimized by taking the inner product with the test function, $\varphi(\xi,\eta) = \ell_i(\xi)\ell_j(\eta)$, yielding

$$\int_{D_k} \left(\tilde{\mathbf{Q}}_t + \nabla \cdot \tilde{\mathbf{F}}\right) \varphi(\xi,\eta)\, \mathrm{d}\boldsymbol{\xi} = 0. \tag{17}$$

Integrating by parts leads to

$$\int_{D_k} \frac{\partial \tilde{\mathbf{Q}}_N}{\partial t} \varphi(\xi,\eta)\, \mathrm{d}\boldsymbol{\xi} + \int_{\partial D_k} \tilde{\mathbf{F}}_N \cdot \mathbf{n}\, \varphi(\xi,\eta)\, \mathrm{d}S - \int_{D_k} \tilde{\mathbf{F}}_N \cdot \nabla \varphi(\xi,\eta)\, \mathrm{d}\boldsymbol{\xi} = 0. \tag{18}$$

To ensure connectivity between each subdomain, we replace $\mathbf{F}_N \cdot \mathbf{n}$ in the boundary integral with a numerical flux $\mathbf{F}^*$, which for conservation laws is typically the solution to an approximate Riemann problem based on $\mathbf{Q}_N$ at the coincident faces from two subdomains. Simplifying (18) with

$$\mathcal{M}_{i,j} = \int_{D_k} \ell_i(\xi)\ell_j(\eta)\, \mathrm{d}\boldsymbol{\xi},$$

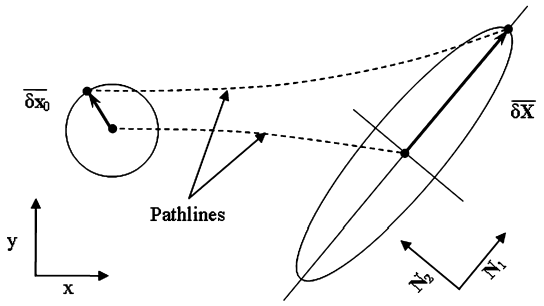


Fig. 3. After a time interval, T, an infinitesimal fluid parcel will be deformed. The orientation of $\overline{\delta\phi}$ is chosen so that it aligns with the eigenvector, $\mathbf{N}_i$, of the right Cauchy–Green strain tensor that is associated with the largest eigenvalue.

$$\mathcal{F}_{i,j} = \int_{\partial D_k} \ell_i(\xi)\ell_j(\eta)\,\mathrm{d}S, \tag{19}$$
$$\mathcal{S}_{i,j} = \int_{D_k} \nabla(\ell_i(\xi)\ell_j(\eta))\,\mathrm{d}\boldsymbol{\xi},$$

we have

$$\mathcal{M}\frac{\partial \tilde{\mathbf{Q}}_N}{\partial t} + \mathcal{F}\tilde{\mathbf{F}}^* - \mathcal{S}\tilde{\mathbf{F}}_N = 0, \tag{20}$$

where $\mathcal{M}$ is the mass matrix and $\mathcal{F}$ is the stiffness matrix.

The remaining system of ordinary differential equations is integrated in time with a 4th-order low-storage, low-dispersion explicit Runge–Kutta method [27].

3. Finite-time Lyapunov exponents (FTLE) and spectral element methods

3.1. FTLE theory

Following Shadden et al. [8], we consider a system of the form

$$\begin{cases} \dot{\mathbf{x}}(t; t_0, \mathbf{x}) = \mathbf{v}(\mathbf{x}(t; t_0, \mathbf{x}_0), t) \\ \mathbf{x}(t_0; t_0, \mathbf{x}_0) = \mathbf{x}_0 \end{cases} \tag{21}$$

where $\mathbf{v}$ is the fluid velocity at point $\mathbf{x}(t; t_0, \mathbf{x}_0)$. A flow map, $\phi_{t_0}^t$, can be defined which maps material points from their initial location $\mathbf{x}_0$ at t_0 to their location at $t = t_0 + T$,

$$\phi_{t_0}^t \equiv \mathbf{x}(t; t_0, \mathbf{x}_0). \tag{22}$$

To reduce notational clutter, we will designate the flow map simply with ϕ hereafter. The differential distance between fluid particles in neighboring trajectories at time t can be expressed in terms of the differential distance between the particles at time t_0 and the deformation gradient

$$\delta\phi = \frac{\partial \phi}{\partial \mathbf{x}_0}\delta\mathbf{x}_0. \tag{23}$$

Squaring both sides gives

$$\|\delta\phi\|^2 = \left\langle \delta\mathbf{x}_0, \frac{\partial \phi}{\partial \mathbf{x}_0}^* \frac{\partial \phi}{\partial \mathbf{x}_0}\delta\mathbf{x}_0 \right\rangle = \langle \delta\mathbf{x}_0, \mathbf{C}\delta\mathbf{x}_0 \rangle, \tag{24}$$

where * represents the matrix transpose and $\mathbf{C}$ is the right Cauchy–Green strain tensor. Since $\mathbf{C}$ is symmetric and positive definite, we can align the basis for $\delta\phi$ with the principle strain axes, i.e. the eigenvectors, $\mathbf{N}$, of $\mathbf{C}$, such that we have $\overline{\delta\phi}$ and $\overline{\delta\mathbf{x}}_0$, as in Fig. 3, and

$$\|\overline{\delta\phi}_i\|^2 = \|\overline{\delta\mathbf{x}}_{0,i}\|^2 \lambda_i(\mathbf{C}), \tag{25}$$

where $\lambda_i(\mathbf{C})$ are the eigenvalues of $\mathbf{C}$. The maximal component of $\overline{\delta\phi}$ is

$$\max \|\overline{\delta\phi}\| = \|\overline{\delta\mathbf{x}}_0\| \sqrt{\lambda_{max}(\mathbf{C})}, \tag{26}$$

where $\lambda_{max}(\mathbf{C})$ is the maximum eigenvalue of $\mathbf{C}$. By assuming that $\max \|\overline{\delta\phi}\|$ grows exponentially in time, we can determine the rate of this growth, or "stretching", by rewriting (26)

$$\max \|\overline{\delta\phi}\| = \|\overline{\delta\mathbf{x}}_0\| \, e^{\sigma |T|}. \tag{27}$$

We define the FTLE (σ) as

$$\sigma(\mathbf{x}_0, T) = \frac{1}{|T|} \ln \sqrt{\lambda_{max} \left(\frac{\partial \phi}{\partial \mathbf{x}_0}^* \frac{\partial \phi}{\partial \mathbf{x}_0} \right)}. \tag{28}$$

Equivalently, we have

$$\sigma(\mathbf{x}_0, T) = \frac{1}{|T|} \ln \left\| \frac{\partial \phi}{\partial \mathbf{x}_0} \right\|_2. \tag{29}$$

According to [5], an LCS is defined as a locally strongest repelling or attracting material line in a flow. However, one must be careful associating ridges in the FTLE field to LCS. For instance, Shadden et al. [8] point out that ridges in the FTLE field do not always ensure zero flux across the material surface associated with that ridge, hence by defining LCSs as ridges in the FTLE field, the LCSs are no longer guaranteed to be Lagrangian. Additionally, ridges in the FTLE field are not strictly hyperbolic, since surfaces of shear stretching are indistinguishable from hyperbolic repelling surfaces in the FTLE field. Nonetheless, as Mathur et al. [28] demonstrate, hyperbolic coherent structures are far more common in real flows of interest than shear structures. Hence, the FTLE field very often provides an accurate picture of LCS in real turbulence. Furthermore, following Farazmand and Haller [6], to identify LCS in two dimensions according to the variational approach of [5] requires a series of criteria to be met on the eigenvector field of the Cauchy–Green strain tensor and the solution to an ordinary differential equation, requiring additional computational costs over the determination of the FTLE field. Hence, identifying LCSs with FTLE fields is more amenable to methods in which the flow map is determined simultaneous to integration of the Navier–Stokes equations over the variational approach.

3.2. Fluid particle tracking algorithm

The particle tracking algorithm requires three steps [29]: 1) the host domain of the particle is located; 2) the fluid velocity is interpolated from the dG grid to the particle's location; 3) the system (21) is integrated in time.

We initialize particles at Chebyshev Gauss–Lobatto quadrature nodes in each subdomain, and numerically integrate the dynamical system (21) over the time interval $T = t - t_0$. The particle final locations form the approximation

$$\phi \approx \bigcup_{k=1}^{K} \Phi_N^k, \tag{30}$$

where Φ_N^k is the Nth-order polynomial approximation of ϕ on the subdomain D_k. Since in dG, the boundaries of conforming subdomains coincide, we remove duplicate boundary particles to improve the efficiency of the particle tracking algorithm.

The particle's host cell is found by comparing the particle coordinate in the unit square to the bounds of the unit square. The inverse of the blending function (4) maps the particle's physical space location to its location, $\boldsymbol{\xi}_p$, within the unit square. For curved sided subdomains, an analytical expression for the inverse map does not exist and so a Newton root-finding method is used to invert the map. A fluid particle leaves a subdomain if it's mapped coordinate is outside the bounds of a unit square. If this is the case, the subdomains shared by the nearest corner determine the new host cell. The fluid velocity is interpolated to the particle location using the existing polynomial approximation of the solution vector, $\mathbf{Q}_N$. Therefore, the interpolation error in the fluid particle velocity is of the same order as the numerical scheme. Particles are advected by integrating (21) in time with an Adams–Bashforth (AB) scheme. The AB is used because it is computationally more efficient as compared to Runge–Kutta schemes used to integrate the Navier–Stokes equations and the linear particle equations are not constrained by numerical stability.

3.3. Forward-time FTLE with spectral methods

The locations of the fluid particles in each subdomain that are traced over the time interval, T, are denoted by Φ_{ij}. These locations construct the interpolant for the flow map given by

$$\Phi_N(\xi, \eta, t) = \sum_{i=0}^{N} \sum_{j=0}^{N} \Phi_{i,j}(t) \ell_i(\xi) \ell_j(\eta). \tag{31}$$

The deformation gradient tensor is determined from Φ_N using the dG operators (15) and (16). Taking into account the mapping to computational space, we compute the components of the deformation gradient tensor at the quadrature nodes as follows,

$$\frac{\partial \Phi}{\partial x_0} = \frac{1}{J} \left[\left(\sum_{k=0}^{N} D_{ik}^{(\xi)} \Phi_{k,j} \right) \frac{\partial y_0}{\partial \eta} - \left(\sum_{k=0}^{N} D_{jk}^{(\eta)} \Phi_{i,k} \right) \frac{\partial y_0}{\partial \xi} \right],$$

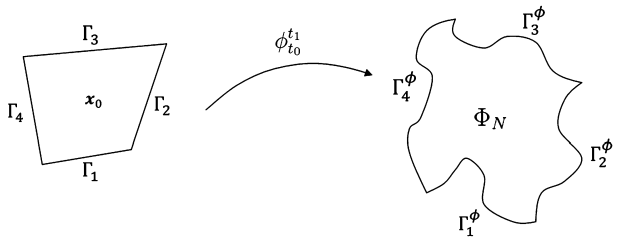


Fig. 4. The nodes that define the flow map subdomain boundaries, Γ_i^{ϕ}, are constructed from the final locations of the nodes initialized at the boundaries of the subdomains at time $t = t_0$ such that $\Gamma_i^{\phi} = \Phi_N|_{\partial\Omega_i}$.

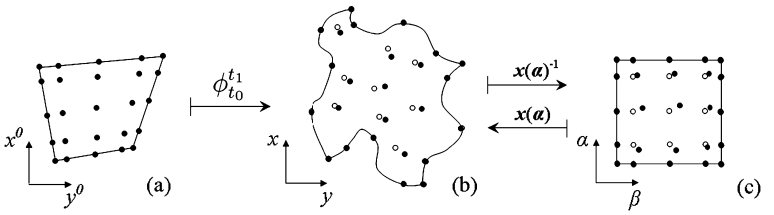


Fig. 5. The fluid tracers are initialized at the Chebyshev–Lobatto quadrature nodes at time $t = t_0$, designated by the filled circles (a). The particles are then advected to time t_1 (filled circles (b)), where they are mapped to the master element with the inverse map $\mathbf{x}(\boldsymbol{\alpha})^{-1}$ (filled circles (c)). The open circles in (b) and (c) correspond to the quadrature nodes in physical space, Φ'_{ij}, and mapped space, $\boldsymbol{\alpha}'_{ij}$, respectively.

$$\frac{\partial \Phi}{\partial y_0} = \frac{1}{J}\left[\left(\sum_{k=0}^{N} D_{jk}^{(\eta)} \Phi_{i,k}\right)\frac{\partial x_0}{\partial \xi} - \left(\sum_{k=0}^{N} D_{ik}^{(\xi)} \Phi_{k,j}\right)\frac{\partial x_0}{\partial \eta}\right]. \tag{32}$$

From the deformation gradient tensor, the FTLE is determined with (29).

3.4. Multiple flow maps from a single particle trace

We can determine the flow map at time $t_2 > t_1$, over time interval $T_2 = t_2 - t_1$,

$$\phi_{t_1}^{t_2} = \mathbf{x}^2(t_2; t_1, \phi_{t_0}^{t_1}), \tag{33}$$

with the same particle trace we used to determine the flow map at t_1 over the time interval $T_1 = t_1 - t_0$,

$$\phi_{t_0}^{t_1} = \mathbf{x}^1(t_1; t_0, \mathbf{x}^0), \tag{34}$$

if we take the flow map at t_1 as the initial state for the flow map at t_2. To do so, we map and interpolate the Nth-order polynomial approximation of the flow map on a subdomain at t_1,

$$\phi_{t_0}^{t_1} \approx \Phi_N, \tag{35}$$

to Gauss quadrature nodes, so that it can be used as the basis for the polynomial approximation of the map at t_2,

$$\phi_{t_1}^{t_2} \approx \Psi_N. \tag{36}$$

Similarly, we find an approximation to the backward-time flow map at t_0,

$$\phi_{t_1}^{t_0} \approx \Theta_N, \tag{37}$$

using the same mapped and interpolated flow map at t_1, (35) as a basis.

3.4.1. Mapping of flow map to unit square

To project the approximation of the flow map at t_1, Φ_N, to the quadrature nodes, the nodes, Φ_{ij}, are first mapped to the master element with the inverse of the linear blending function (4). We parameterize the boundary of the map, Γ_i^{ϕ}, in (4) with

$$\Gamma_i^{\phi} = \Phi_N|_{\partial D_k^i}, \tag{38}$$

i.e. the polynomial approximation of the flow map, Φ_N, at the edges of the subdomains, D_k, as shown in Fig. 4.

The particles that are initialized at the quadrature nodes at time t_0 (Fig. 5(a)), and that are advected until time t_1, form the flow map, Φ_N based on the particle nodes, Φ_{ij} (filled circles, Fig. 5(b)). Through the inverse transfinite map function (4),

Φ_{ij}, maps to the nodes in the unit square, $\boldsymbol{\alpha}_{ij}$ (filled circles, Fig. 5(c)), where we invert the non-linear transfinite mapping function (4) with an iterative Newton method using the quadrature nodes as an initial guess.

The nodes $\boldsymbol{\alpha}_{ij}$ do not necessarily coincide with the quadrature nodes, $\boldsymbol{\alpha}'_{ij}$ (open circles, Fig. 5(c)), i.e.,

$$\boldsymbol{\alpha}_{ij} \neq \boldsymbol{\alpha}'_{ij}, \tag{39}$$

except along the parameterized curves that bound the subdomain, and in the trivial case of uniform flow.

3.4.2. Orthogonal basis for the integrated flow map

On the unit square, we formulate the following interpolant,

$$\Phi_N(\alpha,\beta) = \sum_{i=0}^{N}\sum_{j=0}^{N} \Phi'_{ij}\ell_i(\alpha)\ell_j(\beta). \tag{40}$$

Here, Φ'_{ij} are values of the flow map at t_1 on the quadrature nodes. They are obtained using the known values of the flow map, Φ_{lm}, at the coordinates (α_l,β_m) in the unit square. So,

$$\Phi_{lm} = \sum_{i=0}^{N}\sum_{j=0}^{N} \Phi'_{ij}\ell_i(\alpha_l)\ell_j(\beta_m), \tag{41}$$

which in matrix–vector form can be expressed as

$$\tilde{\Phi}_q = \tilde{\Phi}'_p \mathcal{I}_{pq}, \tag{42}$$

where $\mathcal{I}_{pq}$ is the $(N+1)^2 \times (N+1)^2$ interpolation matrix,

$$\mathcal{I}_{pq} = \ell_i(\alpha_l)\ell_j(\beta_m), \tag{43}$$

and Φ_q and Φ'_p are contiguously aligned, so that $p = i(N+1)+j$ and $q = l(N+1)+m$. Finally, by inverting (42) we have

$$\tilde{\Phi}'_p = \mathcal{I}^{-1}_{pq}\tilde{\Phi}_q. \tag{44}$$

The flow map, $\tilde{\Phi}'_p$ is based an orthogonal, tensorial, quadrature grid and forms the basis at t_1 for new flow maps in forward and backward time.

3.4.3. Construction of backward-time flow maps

We find the backward-time flow map at t_0 by integrating the known flow map at t_1 to t_0,

$$\Theta_{ij} = \Phi_{ij} + \int_{t_1}^{t_0} \left(\frac{\mathrm{d}\Phi_{ij}}{\mathrm{d}t}\right) \mathrm{d}t \tag{45}$$

Since (45) back-traces the particles that we initiate at $\mathbf{x}_0$ to obtain Φ_N in forward time, it follows that Θ_{ij} represent the nodes on the initial dG-mesh, $\mathbf{x}^0_{ij}$ (Fig. 6(b)).

With respect to the flow map, Φ'_{ij}, on the orthogonal quadrature nodes at t_1, we find the flow map in backward time at t_0 (Fig. 6(c)) as

$$\tilde{\Theta}'_p = \mathcal{I}^{-1}_{pq}\tilde{\Theta}_q, \tag{46}$$

according to the definitions in (42) and (43).

We prove (46) by multiplying (45), recast in matrix–vector form, with $\mathcal{I}^{-1}_{pq}$,

$$\mathcal{I}^{-1}_{pq}\tilde{\Theta}_q = \mathcal{I}^{-1}_{pq}\tilde{\Phi}_q + \mathcal{I}^{-1}_{pq}\int_{t_1}^{t_0} \left(\frac{\mathrm{d}\tilde{\Phi}_q}{\mathrm{d}t}\right) \mathrm{d}t. \tag{47}$$

Since the operators are linear we can rewrite this as

$$\mathcal{I}^{-1}_{pq}\tilde{\Theta}_q = \mathcal{I}^{-1}_{pq}\tilde{\Phi}_q + \int_{t_1}^{t_0} \left(\frac{\mathrm{d}\mathcal{I}^{-1}_{pq}\tilde{\Phi}_q}{\mathrm{d}t}\right) \mathrm{d}t. \tag{48}$$

With (44) and (46) it follows that

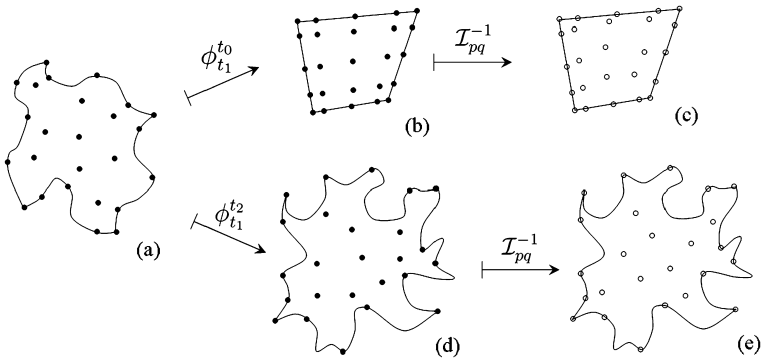


Fig. 6. With the fluid tracers integrated from time t_0 (b) to t_1 (a), then to t_2 (d), we can find the locations of the quadrature nodes at t_2 (e) and t_0 (c) that map from the master element to the flow map at t_1 using interpolation if we invert the interpolation matrix $\mathcal{I}_{pq}$.

$$\mathcal{I}_{pq}^{-1}\tilde{\Theta}_q = \tilde{\Phi}'_p + \int_{t_1}^{t_0} \left(\frac{\mathrm{d}\tilde{\Phi}'_p}{\mathrm{d}t} \right) \mathrm{d}t. \tag{49}$$

The right-hand side represents the integration of orthogonal flow map at t_1 to t_0 and thus is equal to $\tilde{\Theta}'_p$. This proves (46).

Since $\tilde{\Theta}'_p$ represent an approximation of the backward-time flow map on the orthogonal quadrature basis, we can use spectral operators like in (32) to determine the deformation gradients from the flow map.

Alternatively, the backward-time FTLE can be computed using a method by Haller and Sapsis [30], who derive an analytical relationship between the largest backward-time FTLE and the smallest forward-time FTLE. This enables one to compute the backward-time FTLE associated with the flow map, $\phi_{t_1}^{t_0}$, from the forward-time flow map, $\phi_{t_0}^{t_1}$, in the following way:

$$\begin{aligned} \sigma(\phi, -T) &= \frac{1}{|T|} \ln \sqrt{\lambda_{max} \left(\frac{\partial \mathbf{x}_0}{\partial \phi}^* \frac{\partial \mathbf{x}_0}{\partial \phi} \right)} \\ &= -\frac{1}{|T|} \ln \sqrt{\lambda_{min} \left(\frac{\partial \phi}{\partial \mathbf{x}_0}^* \frac{\partial \phi}{\partial \mathbf{x}_0} \right)}. \end{aligned} \tag{50}$$

The backward-time FTLE computed from (50) is determined on the same deformed grid at t_1 that we use for our dG-FTLE algorithm, and so suffers the same computational challenge as the present work. Although (50) presents an efficient and accurate method for computing the backward-time FTLE field from the forward-time flow map, the method Haller and Sapsis [30] present does not allow for the determination of the backward-time flow map, whereas our dG-based method does. Work by Karrasch et al. [31] implies that determination of the backward-time flow map will reduce the computational costs and increase the numerical stability associated with the identification of repelling hyperbolic LCS according to the geodesic theory by Haller [5]. Hence, extensions of this algorithm to identify LCS from geodesic theory will benefit from the high-order accurate backward-time flow map computed from the forward-time flow map afforded by this algorithm. We compare our dG-based approach with the approach by Haller and Sapsis on a vortex test case in Section 4.

3.4.4. *Multiple forward-time flow maps from a single particle trace*

Similar to the backward-time flow map, we find new flow maps in forward-time at time $t_2 > t_1$ with

$$\Psi_{ij} = \Phi_{ij} + \int_{t_1}^{t_2} \left(\frac{\mathrm{d}\Phi_{ij}}{\mathrm{d}t} \right) \mathrm{d}t, \tag{51}$$

from a particle released at the single initial time, t_0. Following Fig. 6(a), (d) and (e), the flow map at t_2 is determined with respect to the flow map, Φ'_{ij}, on the orthogonal quadrature nodes at t_1, as

$$\tilde{\Psi}'_p = \mathcal{I}_{pq}^{-1} \tilde{\Psi}_q, \tag{52}$$

according to the definitions in (42) and (43).

We can reiterate the projection to an orthogonal basis at t_2 and find as many flow maps as we desire from a single initial time particle trace and continue to use consistent spectral differentiation to find the deformation gradient tensor and its corresponding FTLE.

3.4.5. Notes on the dG-FTLE from single particle trace

Pertaining to the use of a single particle trace to determine multiple forward and backward FTLEs, we make the following remarks:

Remark 1. As the flow map time interval increases, gradients in Φ_N increase and hyperbolicity in the flow field causes the flow map topology to become increasingly tangled. This translates to Gibbs oscillations and aliasing errors in the approximation of the flow map.

Remark 2. The distance between the quadrature nodes, $\boldsymbol{\alpha}'_{ij}$, and the inverse-mapped nodes, $\boldsymbol{\alpha}_{ij}$, increases as the flow map gradients increase with time. Since the Newton method uses $\boldsymbol{\alpha}'_{ij}$ as an initial guess, increasing T will affect the convergence of the inverse map.

Remark 3. It is well known that deviations from near-optimal nodal points, such as the Chebyshev Gauss–Lobatto nodes, increase the Lebesgue constant, $\Lambda_N(\mathcal{I})$ [32], which is connected to poor conditioning of the interpolation. Therefore, since $\mathcal{I}$ depends on $\boldsymbol{\alpha}'_{ij}$ as in (43), as the time interval is increased, $\Lambda_N(\mathcal{I})$, the conditioning and interpolation errors increase. As the polynomial order, N, increases, $\Lambda_N(\mathcal{I})$ further increases [32].

Remark 4. $\mathcal{I}_{pq}$ is easily inverted with

$$\mathcal{I}_{pq}^{-1} = [\ell_i(\alpha_l)\ell_j(\beta_m)]^{-1} = \ell_l(\alpha'_i)\ell_m(\beta'_j), \tag{53}$$

where p and q are the same as in (43).

While the conditioning can lead to large interpolation errors, it also offers insights into the deformation of the fluid that compliment the FTLE. So, in addition to the FTLE, we have more measures for the deformation of the subdomains, including: 1) the Lebesgue constant, Λ_N; 2) the condition number, κ, of $\mathcal{I}$, given by

$$\kappa(\mathcal{I}) \equiv \frac{\lambda_{max}(\mathcal{I})}{\lambda_{min}(\mathcal{I})}; \tag{54}$$

and 3) the separation $|\boldsymbol{\alpha}_{ij} - \boldsymbol{\alpha}'_{ij}|$, which we reduce to a single parameter $\Delta\alpha$ defined by

$$\Delta\alpha \equiv \max|\boldsymbol{\alpha}_{ij} - \boldsymbol{\alpha}'_{ij}|. \tag{55}$$

In general, the Lebesgue constant is difficult to determine for an arbitrary two-dimensional set of interpolation nodes. Hence, we do not include Λ_N in our analysis. In the vortex and cylinder test cases, we illustrate the correlation between $\kappa(\mathcal{I})$ and $\Delta\alpha$ and the FTLE.

3.5. Algorithm summary

The algorithm that determines several FTLEs from a single particle trace is summarized as follows:

A. Particle tracing algorithm in dG:
   1. Initialize fluid tracers on Chebyshev Gauss–Lobatto nodes.
   2. Remove double particles at subdomain boundaries.
   3. The velocity solution of a dG Navier–Stokes solver is interpolated to each particle's location.
   4. The particles are advected one step in time using a 3rd-order Adams–Bashforth scheme.
   5. Particle search algorithm:
      i. The particle is mapped to its last known host domain.
      ii. If the particle leaves the host domain, a new host domain is found from the host domains connected to the nearest node.

B. Forward-time FTLE using dG at time t_1:
   1. The deformation gradient tensor is determined using spectral operators on the flow map Φ'_{ij}.
   2. The Cauchy–Green deformation tensor is constructed.
   3. The FTLE is computed according to (28).

C. The FTLE at time t_2:
   1. Given the nodes of the flow map approximations, Φ_{ij} and Ψ_{ij}, find the node locations $\boldsymbol{\alpha}_{ij}$ using the inverse of (4).
   2. Construct the interpolation operator matrix $\mathcal{I}_{pq}$ from the nodes $\boldsymbol{\alpha}_{ij}$ and the quadrature nodes $\boldsymbol{\alpha}'_{ij}$.
   3. Construct the vector $\tilde{\Psi}_q$ from Ψ_{ij} and compute $\tilde{\Psi}'_p$ from (52).
   4. Once Ψ'_{ij} is computed, the FTLE can be computed using (32) and (28).

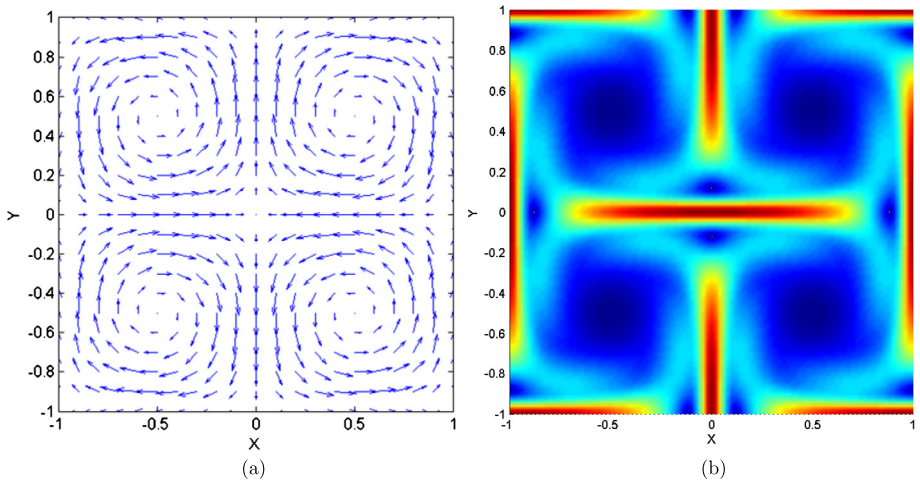


Fig. 7. Velocity field and FTLE field for the periodic gyre flow.

D. Backward-time FTLE:
1. Given the interpolation matrix $\mathcal{I}_{pq}$, construct the vector $\tilde{\Theta}_q$ from Θ_{ij} and compute $\tilde{\Theta}'_p$ from (46).
2. Once Θ'_{ij} is computed, the FTLE is computed using (32) and (28).

4. Test cases

We test the FTLE algorithm for three cases; a steady, spatially periodic gyre flow with a prescribed velocity field that is not a solution to the Navier–Stokes equations; a vortex advected by a uniform, inviscid flow, with a velocity field that is computed with a dG method; and the complex, viscous flow around a square cylinder.

4.1. Gyre flow

The gyre flow is defined by the following velocity field [8]:

$$u(x, y, t) = -\pi A \sin(\pi f(x, t)) \cos(\pi y)$$
$$v(x, y, t) = \pi A \cos(\pi f(x, t)) \sin(\pi y) \frac{\partial f}{\partial x}, \tag{56}$$

where we have

$$f(x, t) = a(t)x^2 + b(t)x, \tag{57}$$
$$a(t) = \epsilon \sin(\omega t), \tag{58}$$
$$b(t) = 1 - 2\epsilon \sin(\omega t). \tag{59}$$

While this is not a solution to the fluid conservation laws, it represents a simplified model of a generic gyre flow, which is a common flow type in geophysical systems. When we set $t = 0$, we have a simplified, time-independent flow given by

$$u(x, y) = -\pi A \sin(\pi x) \cos(\pi y),$$
$$v(x, y) = \pi A \sin(\pi y) \cos(\pi x). \tag{60}$$

As seen in Fig. 7(a), the time-independent gyre velocity field consists of spatially periodic rotating square cells that are one unit wide. The FTLE field (Fig. 7(b)) is a spatially periodic arrangement of ridges bounding the rotating cells seen in the vector field. As the time interval is increased, the gradient around the ridges increases and hence the ridges become sharper.

We determine the FTLE field on two grids for this case; one uniform grid (Fig. 8(a)) and a grid with increased resolution at the locations of the ridges (Fig. 8(b)) using a cosine distribution.

As a reference, we consider a highly resolved computation based on a cosine grid with 32×32 subdomains, a polynomial order of 12, and a time step, $\Delta t = 0.00125$. Coarser computations on a 16×16 grid, with $\Delta t = 0.0025$ and polynomial orders from $P = 2$ to $P = 7$ are compared to the reference case to determine the errors. The FTLE is determined at $T = 2$, 3, and 4 for each case.

Fig. 9 shows that the flow map Φ_N converges spectrally to the fine grid solution at time t_1, while, as T increases, the error increases. The forward-time FTLE at $t_2 = t_0 + 2T$ and the backward-time FTLE computed from the same trace converge

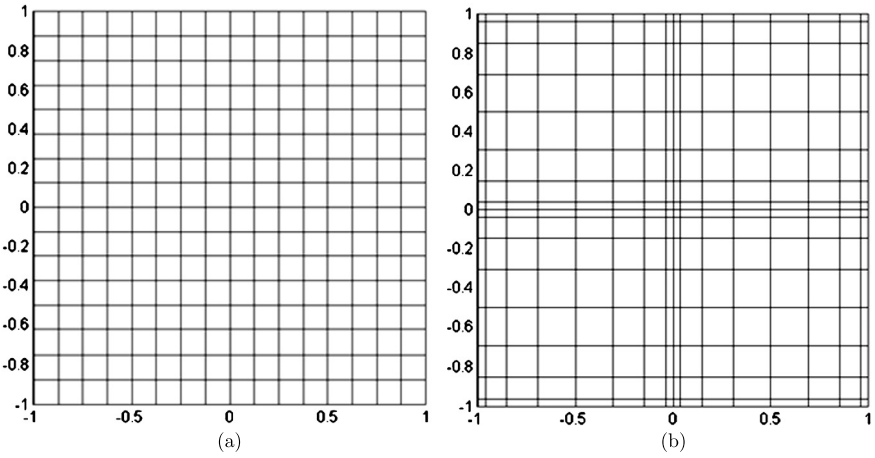


Fig. 8. Uniform distribution of subdomains (a) and a cosine distribution where the subdomains are clustered near the FTLE ridges (b).

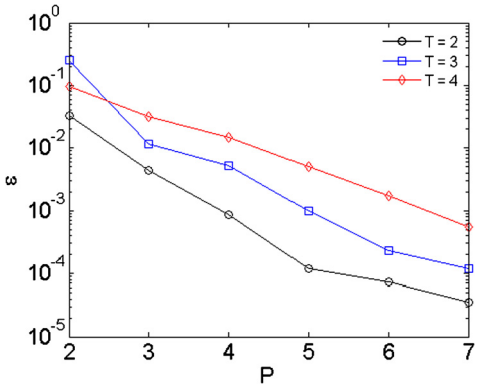


Fig. 9. Convergence for the FTLE fields computed from Φ_N at time intervals $T = 2$, 3, and 4.

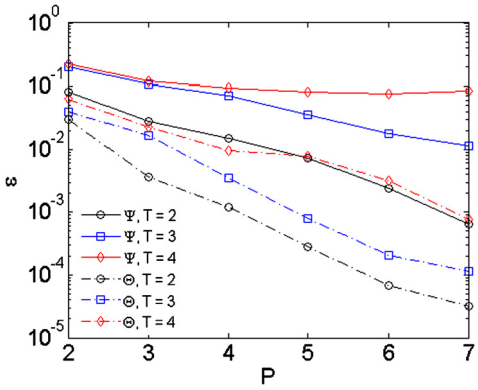


Fig. 10. Convergence for the FTLE fields computed from Ψ_N (forward-time) and Θ_N (backward-time) at time intervals $T = 2$, 3, and 4.

spectrally as well (Fig. 10). Since the gradients are larger in Ψ_N as compared to Θ_N, the error in the deformation gradient tensor increases and hence the error in the forward-time FTLE is larger than in the backward-time FTLE.

The time-dependent gyre flow provides a quantitative test of the accuracy of the algorithm on a simple FTLE field with an LCS that is not grid-aligned, in contrast to the FTLE field for the steady gyre case. The unsteady gyre flow is given by $u(x, y, t)$ and $v(x, y, t)$ in (56) and is a time-periodic perturbation in the x-direction of the steady gyre flow. Time-dependence of the flow field leads to a curved FTLE ridge separating the gyres (Fig. 11). In this study, we set $\epsilon = 0.1$, $A = 0.1$, and $\omega = 2\pi/10$. The FTLE is computed over the domain $[0, 2] \times [0, 1]$. The convergence of the error for the FTLE field computed over time intervals of $T = 3$ and 4 is exponential as depicted in Fig. 12.

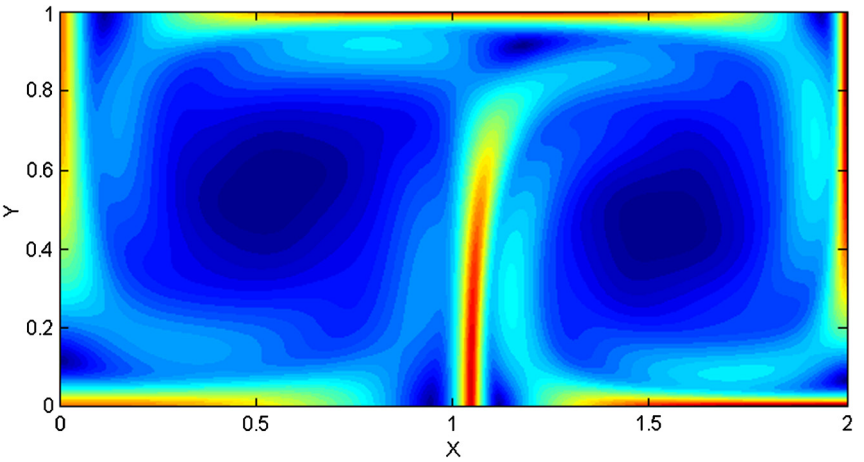


Fig. 11. Time-dependent gyre FTLE field for a time interval of $T = 4$.

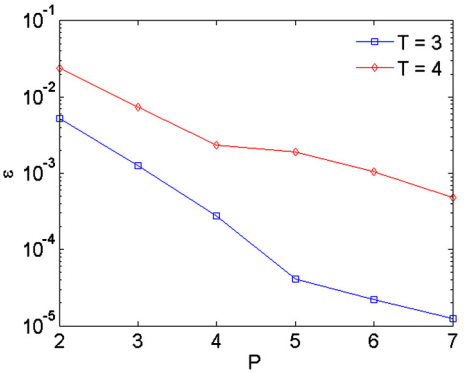


Fig. 12. Convergence for the FTLE fields computed from Φ_N at time intervals $T = 3$ and 4 for the time-dependent gyre flow.

4.2. Vortex advected by uniform flow

We consider the benchmark problem of a circular, isentropic vortex that is advected by a uniform flow. Following [33], the velocity field is given by

$$u(x,y,t) = U_\infty - \frac{A}{R}(y - Y_c)\exp\left(-\frac{(x - X_c - U_\infty t)^2 + (y - Y_c)^2}{2R^2}\right),$$
$$v(x,y,t) = \frac{A}{R}(x - X_c - U_\infty t)\exp\left(-\frac{(x - X_c - U_\infty t)^2 + (y - Y_c)^2}{2R^2}\right), \tag{61}$$

and the temperature field by

$$T = 1 - \frac{1}{2}(AM_\infty)^2 \exp\left(-\frac{(x - X_c - U_\infty t)^2 + (y - Y_c)^2}{R^2}\right)(\gamma - 1), \tag{62}$$

where X_c and Y_c are the initial coordinates of the vortex center. The initial conditions given by (61) and (62) are solutions to the Euler equations. The velocity field at initial time t_0 with the vortex center located at $(X_c, Y_c) = (0.5, 0.5)$ is shown in Fig. 13(a). The uniform free-stream velocity is set to $U_\infty = 1$ and the radius is $R = 0.1$. The vortex strength A is set to 0.2.

The flow map and FTLE field for this velocity field can be determined analytically, as mentioned in [19]. We present the solution in Appendix A.

4.2.1. Errors in the forward-time flow map

The analytical solution to the vortex case provides a reference for a detailed analysis of the numerical errors. There are a number of sources of error in the numerical algorithm to compute the flow map, Φ_N, including the error in the dG computation of the velocity field, and the error in the flow map, Φ_N, with contributions from the numerical integration of the fluid tracers, the grid spacing, and the interpolation error, which in turn is affected by high gradients in the flow map. For a detailed discussion of the fluid particle tracking algorithms and accuracy and convergence of dG methods,

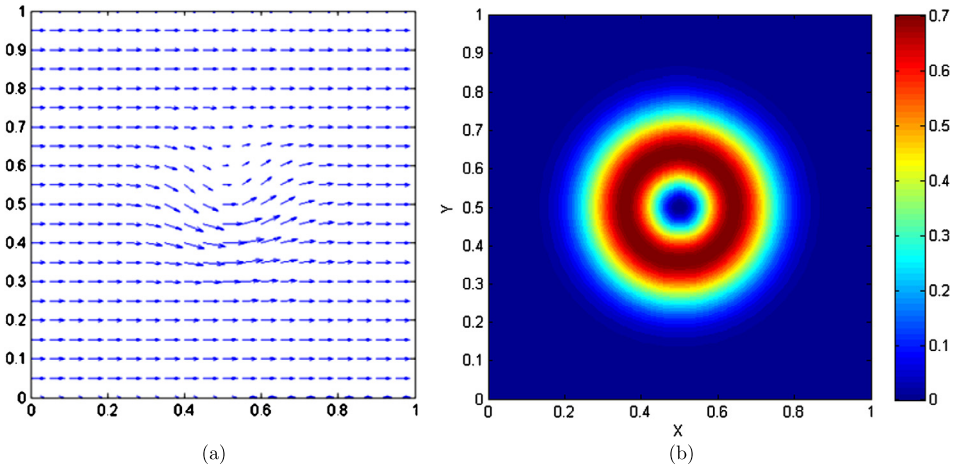


Fig. 13. Velocity field for the periodic vortex flow (a) and the FTLE field for $T = 0.5$ (b).

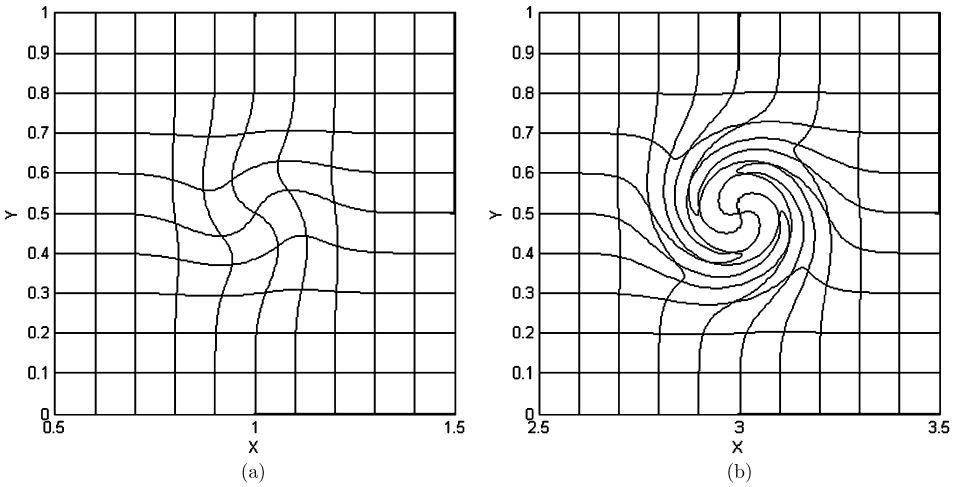


Fig. 14. Subdomain deformation at $T = 0.5$ (a) and $T = 2.5$ (b).

we refer to [29] and [34], respectively. In this work, we ensure that the numerical errors from the dG method and fluid particle integration algorithm are small compared to the flow map error so that the error in the FTLE determination can be examined. In order to quantify the local error of the flow map, we compute the interpolation nodes exactly and compute the difference between the exact flow map, ϕ, and the polynomial approximation, Φ_N, given by

$$\epsilon_\phi = |\Phi_N - \phi|. \tag{63}$$

As the integration time, T, increases, the flow map topology becomes more distorted and tangled (Fig. 14), leading to large gradients. Therefore, the interpolation error must increase with increasing T. The root-mean-square error, ϵ_{RMS}, of Φ_N is given by

$$\epsilon_{RMS} = \sqrt{\frac{1}{M}\left(\sum_{i=1}^{M} |\Phi_N - \phi|^2\right)}, \tag{64}$$

where M is the number of interpolation points. A comparison of ϵ_{RMS} computed at $T = 0.5$ and $T = 5$ is plotted in Fig. 15 versus the polynomial order, P. Spectral accuracy is seen in Φ_N for both T and with increasing T, ϵ_{RMS} increases significantly.

4.2.2. Errors in the flow map interpolation

In addition to the error introduced by the flow map, Φ_N, the errors in the new flow maps Ψ_N and Θ_N are affected by the conditioning of the interpolation matrix $\mathcal{I}$. In Fig. 16, the maximum condition number, $\kappa(\mathcal{I})$, and the maximum value of the parameter $\Delta\alpha$ are plotted as a function of time for a 4th-order and 6th-order solution. As Fig. 16 illustrates, κ increases as the time interval increases, as does $\Delta\alpha$. In both cases, a reduction in the magnitude of the parameters is observed when

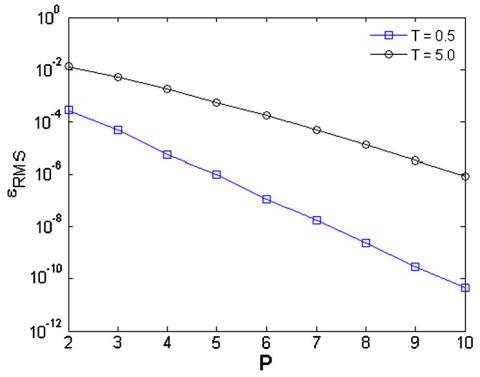


Fig. 15. The convergence of the RMS error of Φ_N at two times, $T = 0.5$ and $T = 5.0$.

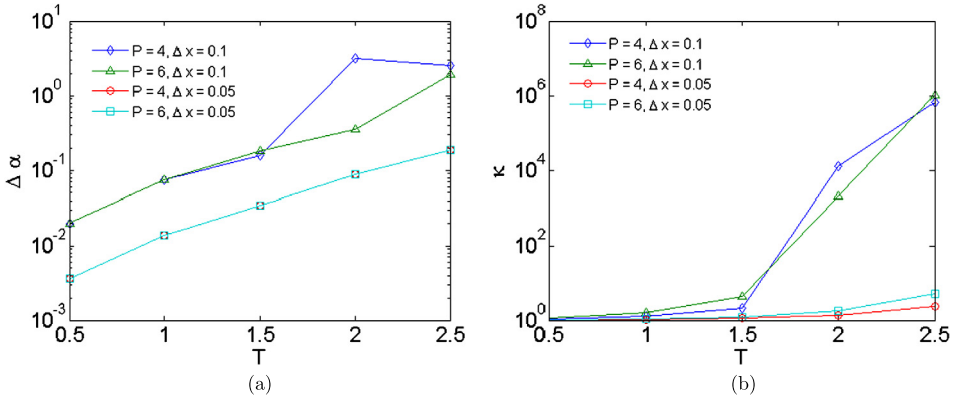


Fig. 16. Dependency of maximum condition number on polynomial order and subdomain deformation. The refined case uses a uniform grid of $\Delta x = 0.05$.

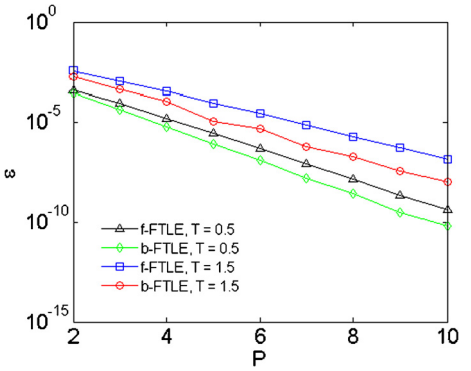


Fig. 17. The convergence rates of the RMS error of the backward-time and forward-time flow maps found using matrix inversion at two time intervals, T.

the grid is refined. The relationship between κ and the Lebesgue constant, Λ_N results in an increase in κ when P increases. However, $\Delta\alpha$ exhibits no dependency on P. This is because $\Delta\alpha$ depends only on the relationship between the topology of the flow map, Φ_N, and the transfinite mapping function, $\mathbf{x}(\boldsymbol{\alpha})$.

Both the forward and backward-time flow maps are spectrally convergent for two different time intervals (Fig. 17). However, the magnitude of the error for the $T = 5$ flow maps is greater than the error for the $T = 0.5$ flow maps. Fig. 18 depicts the convergence of the FTLE computation from the flow maps Ψ_N and Θ_N with a time interval $T = 0.5$. Given that Ψ_N and Θ_N converge spectrally, their respective FTLEs converge spectrally since spectral operators are used to compute the deformation gradient tensor.

The backward-time FTLE from time interval $[t_1, t_0]$ computed using the method presented by Haller and Sapsis [30] is compared to the method presented in this work and the convergence of ϵ_{RMS} is depicted in Fig. 19. The error in the method presented in this work compares well with the error in the Haller–Sapsis method. Differences arise from errors due to the

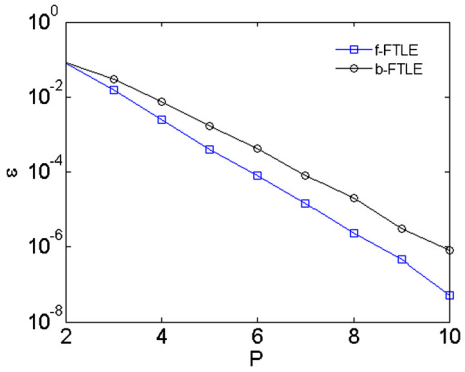


Fig. 18. The convergence rates of the backward and forward FTLE fields at time interval $T = 0.5$.

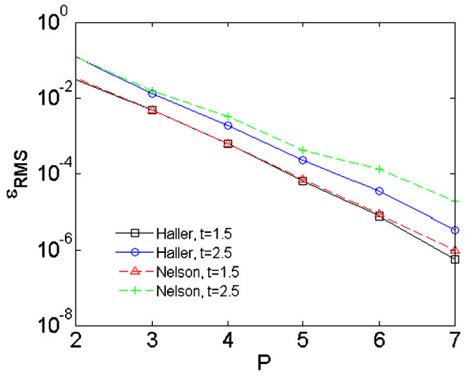


Fig. 19. The error in the backward-time FTLE computed at two time intervals — $T = 1.5$ and $T = 2.5$ — using the method in Haller and Sapsis [30] and this work.

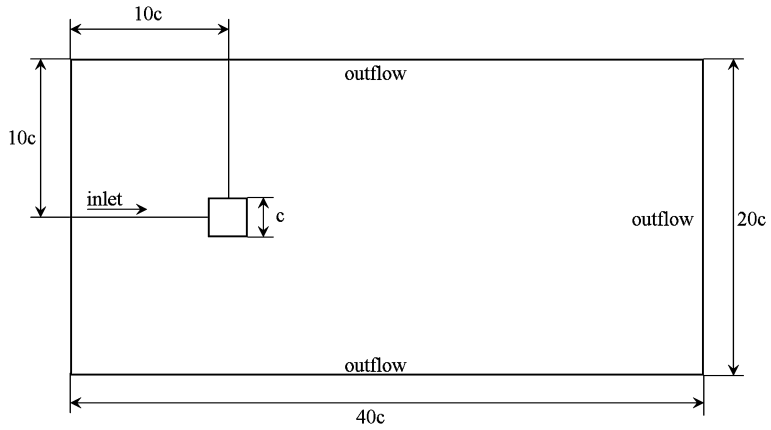


Fig. 20. Problem setup for square cylinder case.

poor conditioning of the interpolation problem at high deformation and high polynomial orders in the method in this work, whereas the Haller–Sapsis method does not require interpolation and, hence, does not suffer from interpolation errors.

4.3. Flow around a square cylinder

We consider the unsteady flow around a square cylinder [10,35] to evaluate the FTLE algorithm on the complex, time-dependent viscous flow around a more complex geometry. The simulation is set up per the schematic in Fig. 20. The reference length of the flow is the cylinder width, $c = 1$. The computational domain size is $40c \times 20c$ with the center of the cylinder located $10c$ from the left boundary and $10c$ from the top and bottom boundaries. Uniform inflow conditions are set on the left boundary, with uniform outflow conditions set at the top, bottom and right. The cylinder wall is adiabatic. The Reynolds number and Mach number based on the free-stream velocity and cylinder width are $Re = 150$ and $M = 0.3$

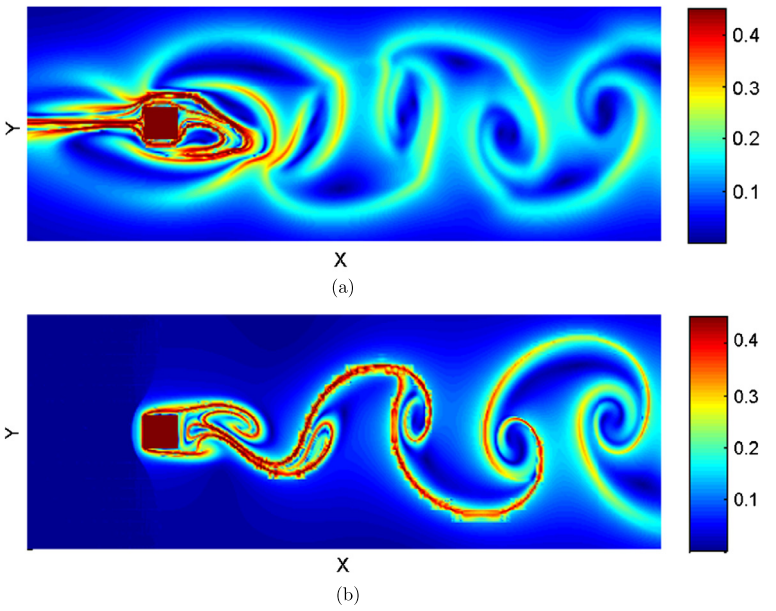


Fig. 21. Comparison of the forward-time FTLE field (a) and the backward-time FTLE field (b). The time interval is $T = 10$.

respectively. The grid is refined until grid independence is achieved. The results presented in this section are based on the refined grid solution. The fluid computation is performed until quasi-steady state is reached [10,35].

A von Karman vortex street develops in the wake of the cylinder, and provides velocity fields that are well suited for study using LCS methods [18]. The forward-time FTLE field exhibits a strong repelling LCS along the stagnation streamline upstream of the cylinder, while the vortex boundaries can be seen in the wake (Fig. 21(a)). In the backward-time FTLE field, strong attractors form in the near wake, where fluid becomes entrained just behind the cylinder in a near-wake recirculation region (Fig. 21(b)). Additional attractors form at the vortex cores and along material lines connecting the vortices. The time interval is $T = 10$, and the FTLE fields are computed using the algorithm presented in [19].

The effect of the conditioning of $\mathcal{I}$ and the errors in the inverse transfinite map is most significant in the regions with the highest gradients in the flow map. This generally correlates to regions with high FTLE values, as seen in Fig. 22. In Fig. 22(b), the log of the condition number, $\ln(\kappa)$, is plotted. The subdomains with the highest ridges near the cylinder and along the stagnation streamline exhibit the largest κ values, which tends to wash out the more subtle increases in κ around the ridges that form in the wake. Alternatively, the parameter, $\Delta\alpha$, in Fig. 22(c), scales in a way that allows improved visualization of the flow structures in the far wake.

Fig. 23 shows the backward-time FTLE determined using the standard approach, in which the fluid tracers are integrated backward in time (Fig. 23(a)), and the backward-time FTLE determined from the forward-time flow map using the algorithm presented in this work. Since the backward-time FTLE in Fig. 23(b) is computed on a highly deformed grid, the values of the FTLE are interpolated onto a uniform grid using cubic interpolation so that the results can be better visualized. The high FTLE values in the left side of Fig. 23(b) are a result of the lack of data in that region since the tracers that were initialized upstream of the cylinder have advected downstream. Both methods agree very well and depict the same stable manifolds in the wake with minor differences in the near wake resulting from the poor conditioning of the subdomains in those areas for Fig. 23(b).

The forward-time FTLE field is depicted in Fig. 24 for the time interval $T = 10$ and at the start times $t_0 = 0$ (24(a)), $t_0 = 1$ (24(b)), and $t_0 = 2$ (24(c)). The frames in Fig. 24 were determined from a single particle integration, initialized at $t = 0$, illustrating the potential for the numerical method presented in this work to compute FTLE fields at different starting states while only integrating one set of particles.

5. Conclusions

An algorithm is developed that computes FTLE fields with dG operators at multiple times, both in backward-time and in forward-time, from a single particle trace. The algorithm is designed to be efficient and consistent in implementation and accuracy with high-order dG methods used to solve the Navier–Stokes equations for the underlying velocity vector fields and can be used simultaneously with the integration of the Navier–Stokes equations, as opposed to post-processing.

By seeding particles on the dG nodal mesh points, the integrated flow maps find their basis in the piecewise representation of the dG mesh, ensuring spectral convergence of deformation gradients and FTLE fields if dG operators are used.

Projection to quadrature on a unit square of integrated flow maps that are initiated on nodal points provides a basis for the determination of forward and backward flow maps at several times from particles that are seeded at a single initial

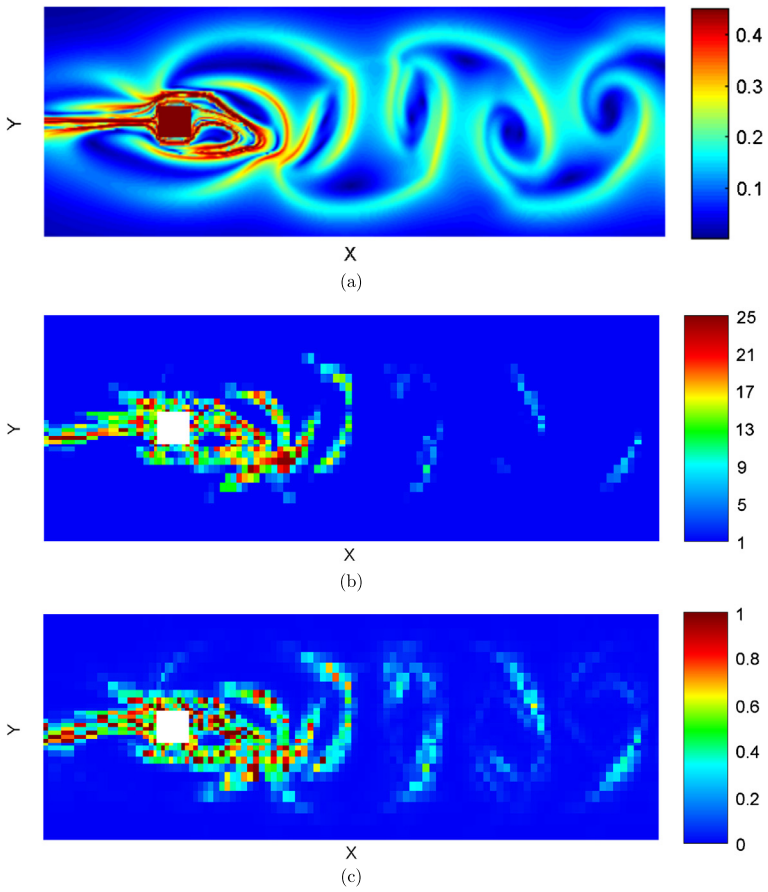


Fig. 22. Comparison of the forward-time FTLE field (a) to the log of the condition number, $\ln(\kappa)$ (b), and the parameter $\Delta\alpha$ (c). The time interval is $T = 10$.

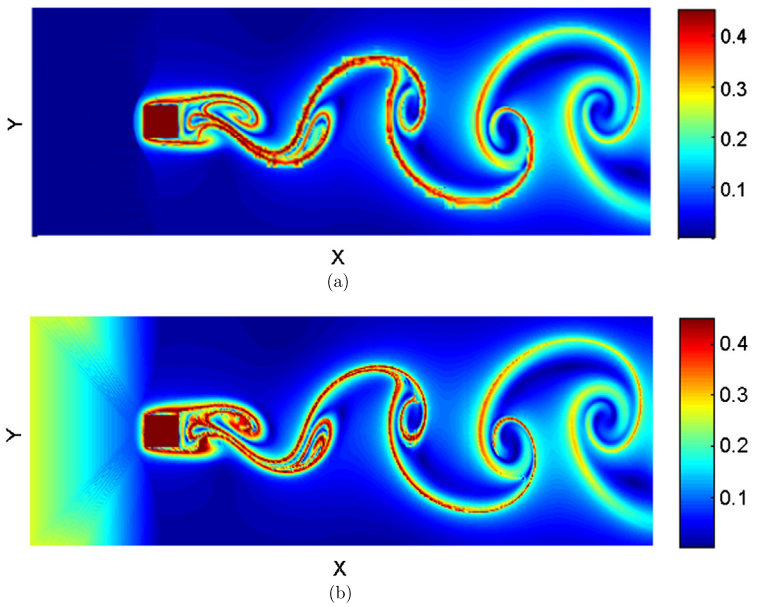


Fig. 23. Comparison of the backward-time FTLE computed from a particle trace in backward-time (a) and the backward-time FTLE computed from the forward-time flow map using Lagrange interpolation (b). The time interval is $T = 10$.

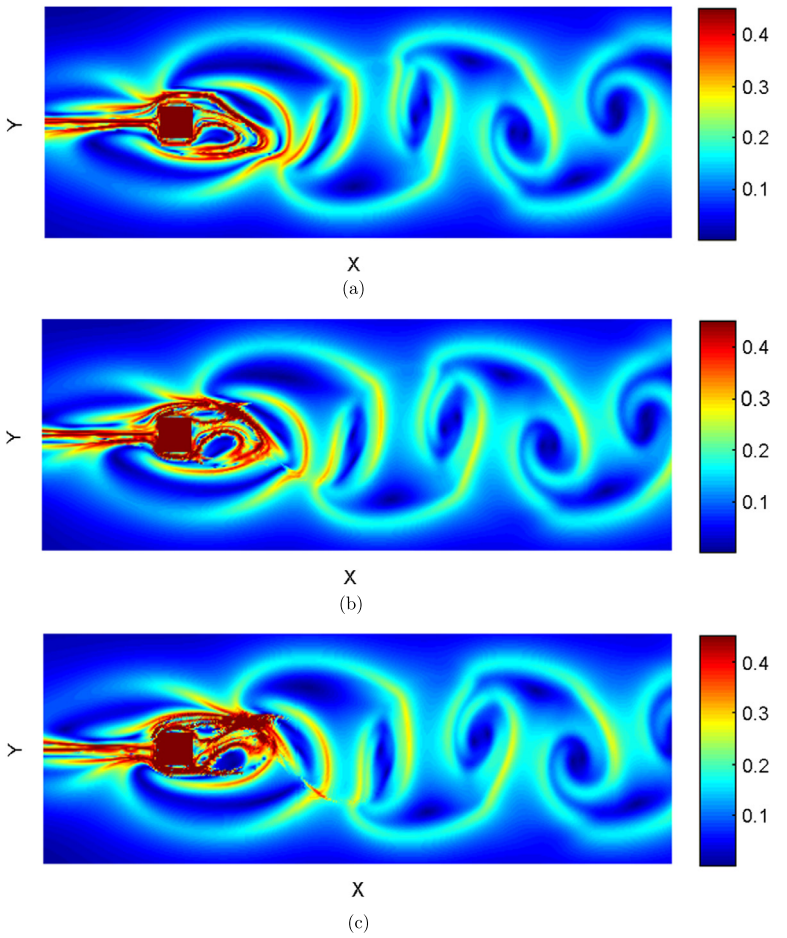


Fig. 24. The forward-time FTLE computed with a time interval $T = 10$ at times $t_0 = 0$ (a), 1 (b), and 2 (c) with one particle trace.

time. Conditioning of the projection is poor when the flow map deforms significantly in time. Quantitative measures for the conditioning also provide additional information regarding the deformation of the flow field.

The algorithm is tested on three benchmarks: a spatially periodic gyre flow; the advection of a vortex in a uniform, inviscid flow; and the unsteady, viscous flow around a square cylinder. Spectral convergence is demonstrated on the gyre case and the vortex case.

The sources for error are examined in detail on the gyre and vortex cases. The most significant source of error is in the interpolation operator, $\mathcal{I}$, where the conditioning for the interpolation depends on the deformation of the flow map. As the time interval, T, increases, the increasing deformation of the subdomains with high flow map gradients leads to poor conditioning for the interpolation to other flow maps, as measured by the condition number, κ, and the parameter $\Delta\alpha$. The parameters κ and $\Delta\alpha$ are shown to highly correlate with the locations of ridges in the FTLE field for the square cylinder case.

The method for computing the backward-time FTLE from the forward-time flow map using dG operators is compared to the method proposed by Haller and Sapsis. The accuracy and convergence of the dG method compare well with the Haller–Sapsis method. The dG method has the advantage of computing the backward-time flow map directly from the forward-time flow map, providing for the ability to analyze the backward-time flow map if needed. The Haller–Sapsis method, on the other hand, determines the backward-time FTLE directly from the forward-time flow map, without determining the backward-time flow map at any point. The dG method, has the drawback that it is more computationally intensive than the Haller–Sapsis method.

The present work focuses on implementation of the algorithm in two dimensions. Extension to three dimensions is straightforward. In future work, we aim to present test cases in three dimensions. The algorithm can be extended to compute the eigenvector field of the Cauchy–Green strain tensor with spectral convergence. Therefore, hyperbolic material surfaces can be extracted with high-order accuracy using the LCS variational theory. We aim to examine the application of this algorithm to the work of Farazmand and Haller [6]. Our future efforts will further focus on developing an adaptive mesh algorithm for improving the conditioning of $\mathcal{I}$ and extending the time interval over which the algorithm can be applied, as well as parallelization, and improvements in efficiency.

Acknowledgements

The authors would like to express their gratitude to Solar Turbines and San Diego State University for their financial contributions to this work, as well as David Kopriva for the provision of his dG code. The authors would also like to thank Sutanu Sarkar for his efforts in co-advising the SDSU–UCSD Joint Doctoral Program.

Appendix A

The flow map for the uniformly advected vortex case with velocity field given by (61) can be computed if we assume a change of coordinates such that

$$x' = x - X_c - U_\infty(t - t_0),$$
$$y' = y - Y_c, \tag{A.1}$$
$$u' = u - U_\infty,$$
$$v' = v. \tag{A.2}$$

Then, the velocity field simplifies to

$$u'(x', y') = -\frac{A}{R} y' \exp\left(\frac{-r^2}{2R^2}\right),$$
$$v'(x', y') = \frac{A}{R} x' \exp\left(\frac{-r^2}{2R^2}\right), \tag{A.3}$$

where $r = \sqrt{(x')^2 + (y')^2}$. By transforming to plane polar coordinates, the velocity can be integrated. The velocity becomes

$$u_r = 0,$$
$$u_\theta = \frac{A}{R} \exp\left(\frac{-r^2}{2R^2}\right) r. \tag{A.4}$$

After integrating with respect to $t - t_0$, and transforming back to Cartesian coordinates, we have the trajectory equations,

$$x(t; t_0, \mathbf{x}_0) = U_\infty(t - t_0) + X_c + r\cos\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)(t - t_0)\right], \tag{A.5}$$

$$y(t; t_0, \mathbf{x}_0) = Y_c + r\sin\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)(t - t_0)\right], \tag{A.6}$$

where, in this case, we note that r is constant for each fluid particle according to Eq. (A.4). The components to the deformation gradient, $\nabla\phi$, are,

$$\frac{\partial\phi^x}{\partial x_0} = \cos\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\frac{x_0 - X_c}{r}$$
$$- r\sin\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\left[-\frac{y_0 - Y_c}{r^2} - \frac{A(x_0 - X_c)}{R^3}\exp\left(-\frac{r^2}{2R^2}\right)t\right], \tag{A.7}$$

$$\frac{\partial\phi^x}{\partial y_0} = \cos\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\frac{y_0 - Y_c}{r}$$
$$- r\sin\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\left[\frac{x_0 - X_c}{r^2} - \frac{A(y_0 - Y_c)}{R^3}\exp\left(-\frac{r^2}{2R^2}\right)t\right], \tag{A.8}$$

$$\frac{\partial\phi^y}{\partial x_0} = \sin\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\frac{x_0 - X_c}{r}$$
$$+ r\cos\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\left[-\frac{y_0 - Y_c}{r^2} - \frac{A(x_0 - X_c)}{R^3}\exp\left(-\frac{r^2}{2R^2}\right)t\right], \tag{A.9}$$

$$\frac{\partial\phi^y}{\partial y_0} = \sin\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\frac{y_0 - Y_c}{r}$$
$$+ r\cos\left[\arctan\left(\frac{y_0 - Y_c}{x_0 - X_c}\right) + \frac{A}{R}\exp\left(-\frac{r^2}{2R^2}\right)t\right]\left[\frac{x_0 - X_c}{r^2} - \frac{A(y_0 - Y_c)}{R^3}\exp\left(-\frac{r^2}{2R^2}\right)t\right]. \tag{A.10}$$

Since $\nabla\phi$ is a 2×2 tensor, the maximal eigenvalue is given explicitly by

$$\lambda_{max}(\mathbf{C}) = \frac{1}{2}\left[\left(\frac{\partial\phi^x}{\partial x_0}^2 + \frac{\partial\phi^y}{\partial y_0}^2\right) + \sqrt{4\left(\frac{\partial\phi^y}{\partial x_0}\frac{\partial\phi^x}{\partial y_0}\right)\left(\frac{\partial\phi^y}{\partial x_0}\frac{\partial\phi^x}{\partial y_0}\right) + \left(\frac{\partial\phi^x}{\partial x_0}^2 - \frac{\partial\phi^y}{\partial y_0}^2\right)^2}\right], \tag{A.11}$$

substituting (A.7)–(A.10). The FTLE is then computed using (28).

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